

Exp No: 14 EMBEDDED C PROGRAM TO DISPLAY "HELLO WORLD" MESSAGE

Github Link: <https://github.com/Pardhav2006/embedded-system>

Program:

```
#include <reg51.h>

#include <stdio.h>

void UART_Init(void)
{
    SCON = 0x50;    // Mode 1, 8-bit UART, enable receiver
    TMOD = 0x20;    // Timer1 Mode2 (8-bit auto reload)
    TH1 = 0xFD;     // 9600 baud rate for 11.0592 MHz crystal
    TR1 = 1;        // Start Timer1
    TI = 1;         // Set TI for first transmission
}

char putchar(char c)
{
    SBUF = c;       // Load character into UART buffer
    while(TI == 0); // Wait for transmission complete
    TI = 0;
    return c;
}

void main(void)
{
    UART_Init();
    while(1)
    {
        printf("Hello World!\r\n");
    }
}
```

Out put:

The screenshot displays the µVision IDE interface. The main window shows the disassembly of a C program. The program is defined in `exp_14.c` and consists of the following code:

```
21: void main(void)
22: {
23:     UART_Init();
24: }

exp_14.c
9   TR1 = 1;      // Start Timer1
10  TI  = 1;      // Set TI for first transmission
11: }
12
13 char putchar(char c)
14 {
15     SBUF = c;    // Load character into UART buffer
16     while(TI == 0); // Wait for transmission complete
17     TI = 0;
18     return c;
19 }
20
21 void main(void)
22 {
23     UART_Init();
24
25     while(1)
26     {
27         printf("Hello World!\r\n");
28     }
29 }
```

The UART window (UART #1) shows the output of the program:

```
Hello World!
Hello World!
Hello World!
Hello World!
Hello World!
```

The status bar at the bottom indicates the simulation is running at 11: 180.34287109 sec, L22 C:2, CAP: NUM SCRL: OVR: R/W.